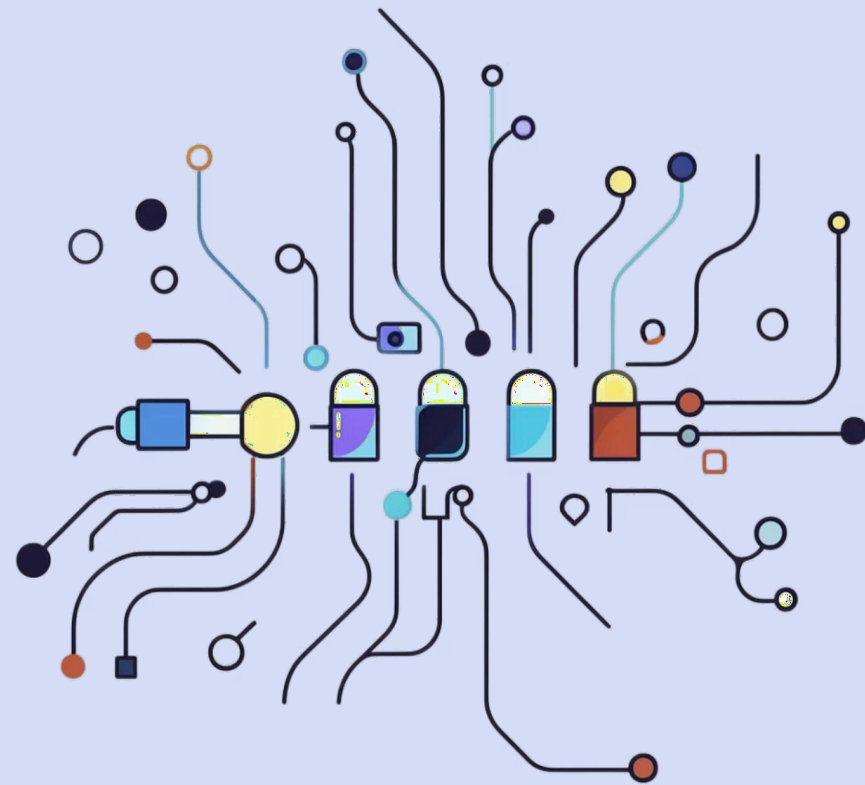




Mini Project : Pulse Generator LED Light

A mini electronics project exploring timing circuits, waveform generation, and LED control.

Introduction

Project Overview

A pulse generator LED light is a simple electronic mini project that produces periodic electrical pulses to control the blinking of an LED.

Widely used in basic electronics learning to understand **timing circuits, waveform generation,** and **switching behavior** of components.

Mechanism & Control

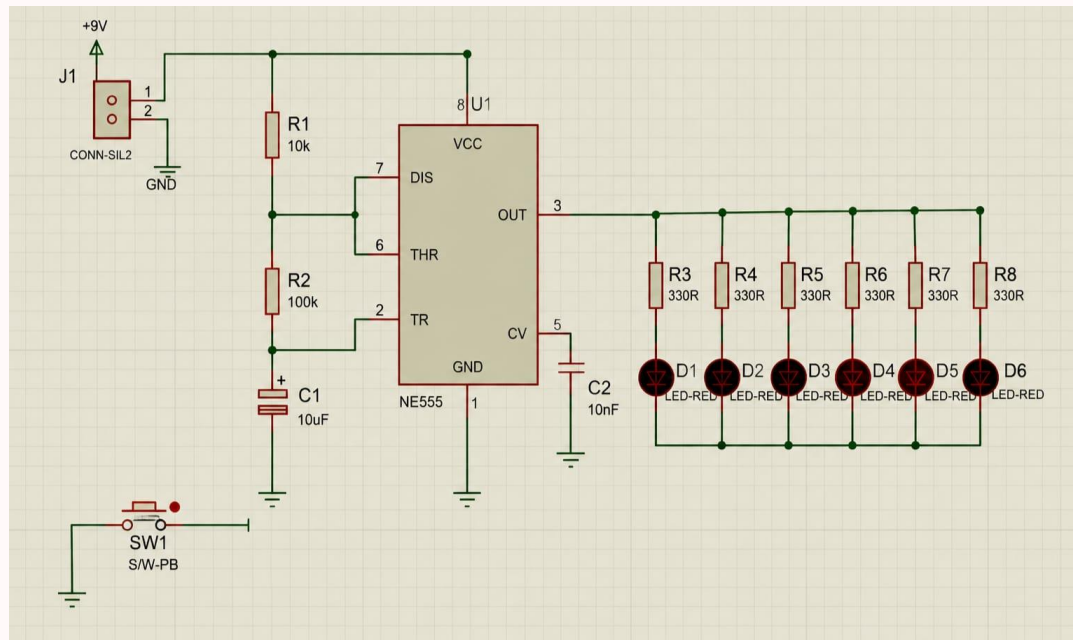
555 Timer in Astable Mode

The circuit uses a **555 Timer IC** operating in astable mode to generate square wave pulses, turning the LED ON and OFF at regular intervals.

Frequency Control

Blinking frequency is precisely controlled by varying **resistor and capacitor values** in the RC network.

Circuit Diagram & Components



CD4017

Decade Counter
IC

555 Timer IC

Pulse generation

LEDs (D1–D6)

High-brightness
Red LEDs

Resistors (R1–
R6)

Current limiting —
 330Ω

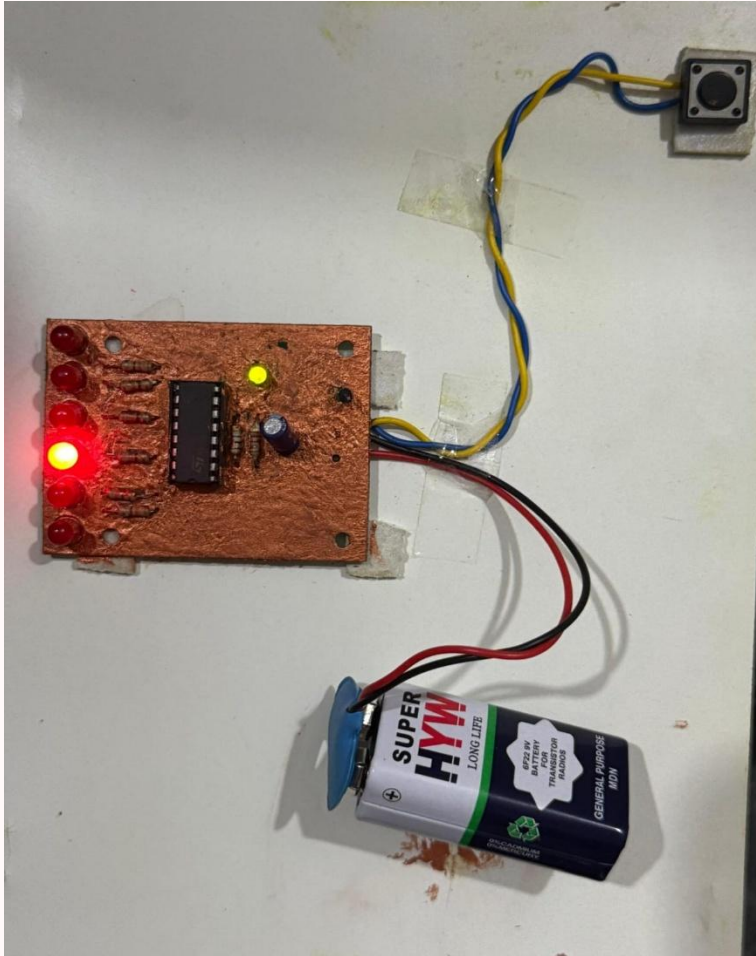
Capacitor

Electrolytic timing
— $10\mu\text{F}$

Power & Base

9V Battery + PCB
Board

PCB Layout



The physical board design ensures structural integrity and reliable signal routing.

Optimized Routing

Copper paths minimize noise

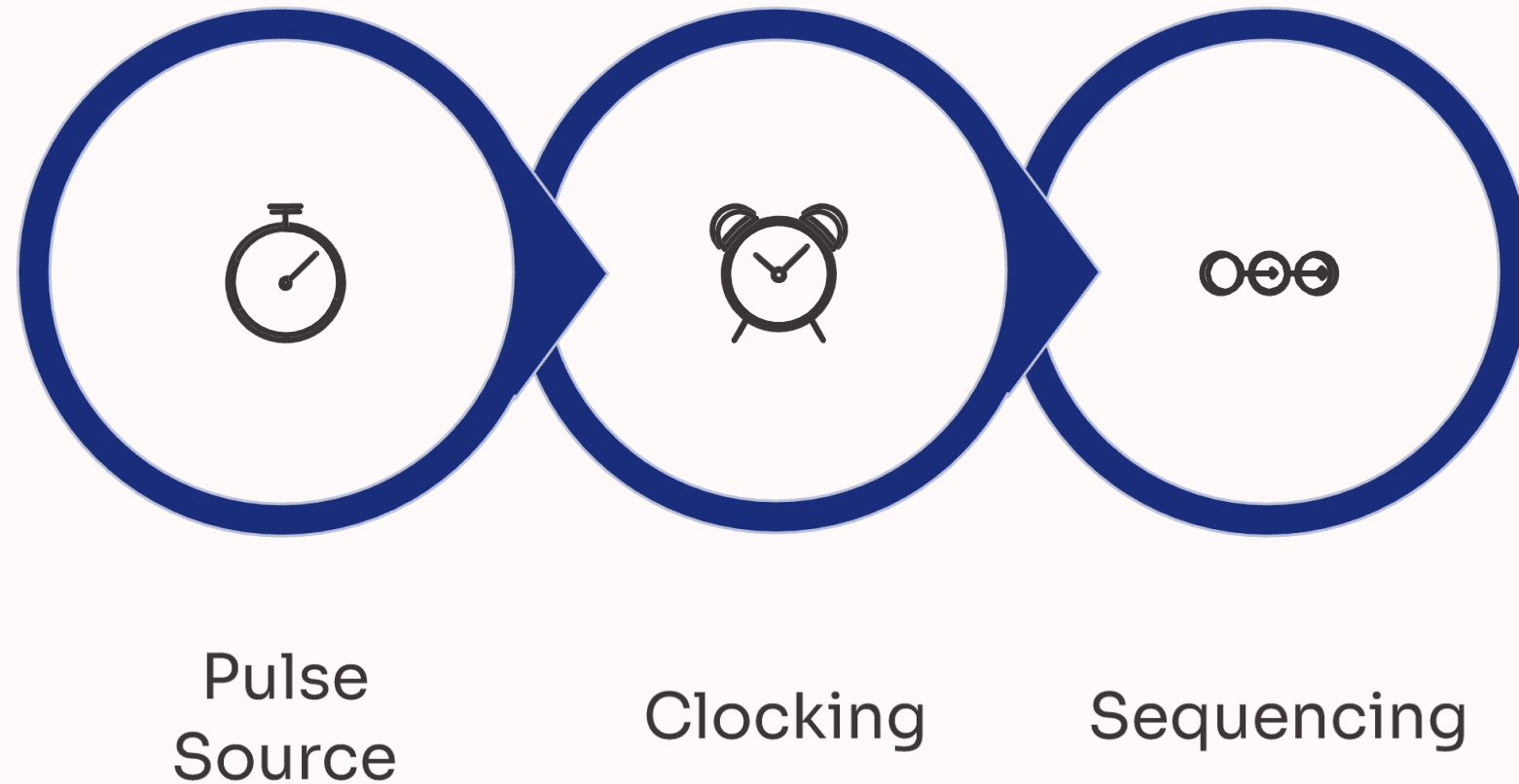
IC Placement

Strategic positioning of 555 and CD4017

LED Alignment

Sequential layout for chaser effect

Functional Working



The 555 timer drives the CD4017 counter, which sequentially activates each LED output — creating a smooth, repeating chaser effect.

Applications



Decorative Lighting

Festive and ornamental LED chaser displays



Automotive Signals

Turn signal and indicator systems



Educational Tools

Hands-on learning for digital logic basics

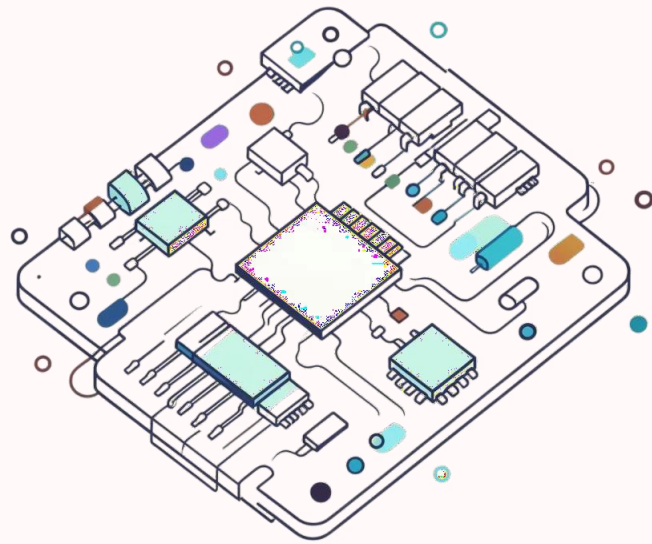


Visual Metronome

Timing reference for rhythm and sequencing



Conclusion



This project illustrates the synergy between **analog timing** and **digital logic** — providing essential hands-on experience in circuit assembly and practical electronics design.

Analog

555 Timer waveform generation

Digital

CD4017 sequential counting

Thank You

Pulse Generator LED Light — Mini Project Presentation

